Computer Vision – Lecture 5

Segmentation

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Course Outline

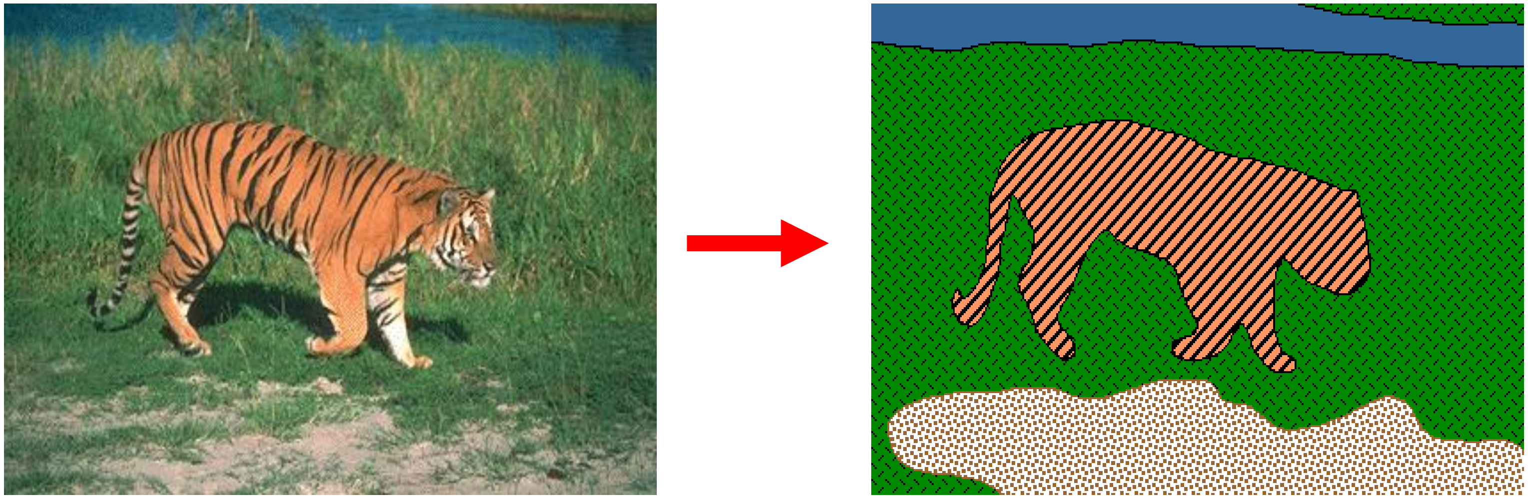
- Image Processing Basics
- Segmentation
  - Segmentation as Pixel-wise Classification
  - Graph-theoretic Segmentation
- Object Categorization
- Deep Learning Approaches for Vision
- 3D Reconstruction

Topics of This Lecture

- Segmentation as pixel-wise classification
  - Probabilistic classification
  - Mixture of Gaussians, EM
- Segmentation as energy minimization
  - Markov Random Fields
  - Energy formulation
- Graph cuts for image segmentation
  - Basic idea
  - s-t Mincut algorithm
  - Extension to non-binary case

Image Segmentation

- Goal: identify groups of pixels that go together



Slide credit: Steve Seitz, Kristen Grauman

How to Approach Segmentation

- We could group pixels using unsupervised clustering
  - Grouping pixels that "look similar"
  - This is the only thing we can do if we do not have annotated data that we could learn from what to segment
  - However, the results are not very good.

  ⇒ *Unsupervised segmentation is very limited,*

- Better approach: Semantic Segmentation
  - Learn a classifier to assign a semantic class C_k to every pixel
  - This requires annotated training data to define what the semantic classes should look like

  ⇒ *Widely used in practice*

  ⇒ *How can we learn such a pixel-wise classifier?*

Feature Space

- Depending on what we choose as the *feature space*, we can classify pixels in different ways.

- Classifying pixels based on intensity similarity

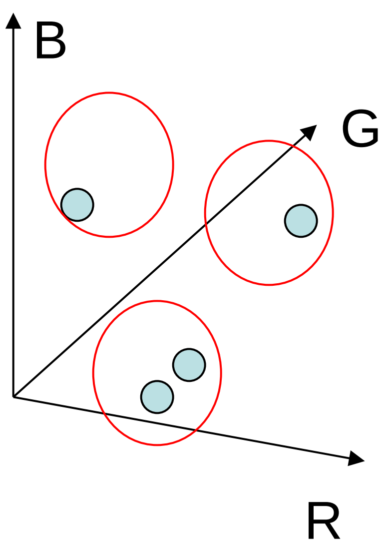
- Feature space: intensity value (1D)

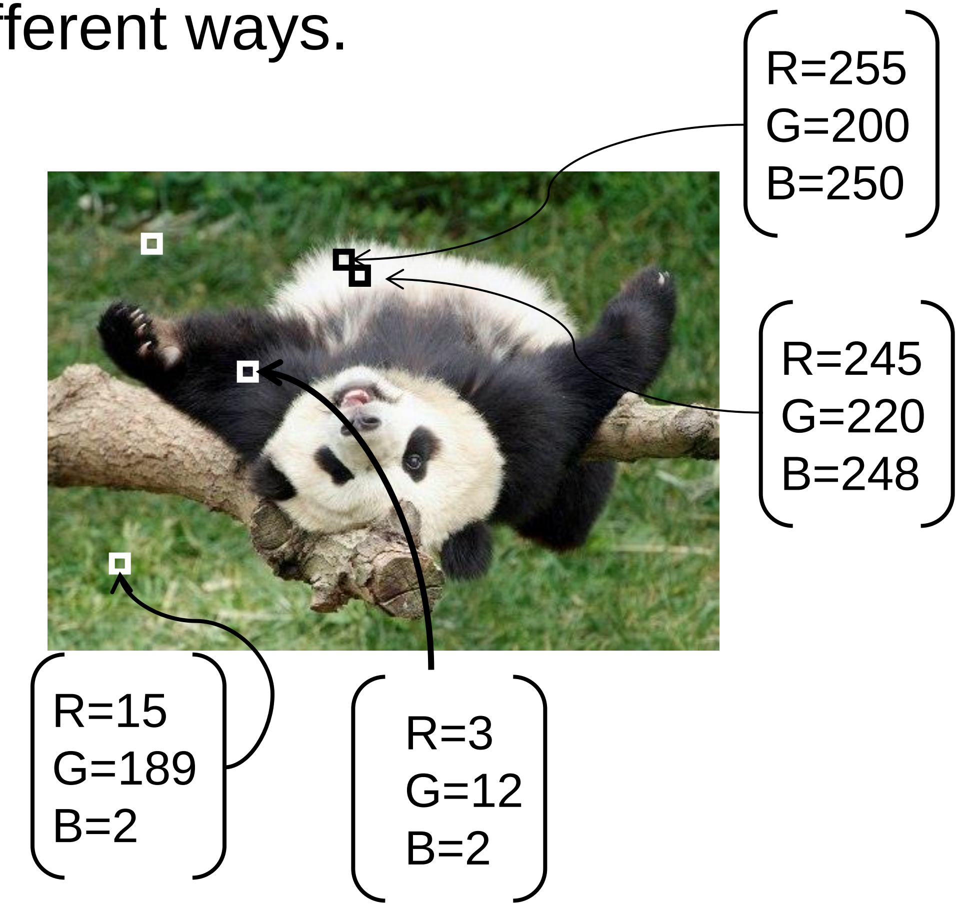
Slide credit: Kristen Grauman

Feature Space

- Depending on what we choose as the *feature space*, we can classify pixels in different ways.

- Classifying pixels based on color similarity





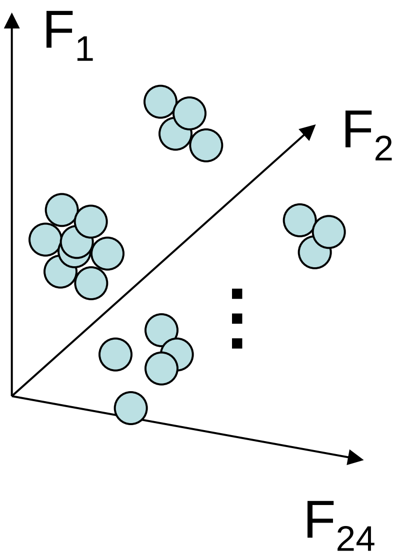
- Feature space: color value (3D)

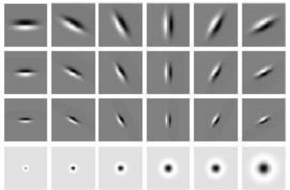
Slide credit: Kristen Grauman

Feature Space

- Depending on what we choose as the *feature space*, we can classify pixels in different ways.

- Classifying pixels based on texture similarity





Filter bank of 24 filters

- Feature space: filter bank responses (e.g., 24D)

Slide credit: Kristen Grauman

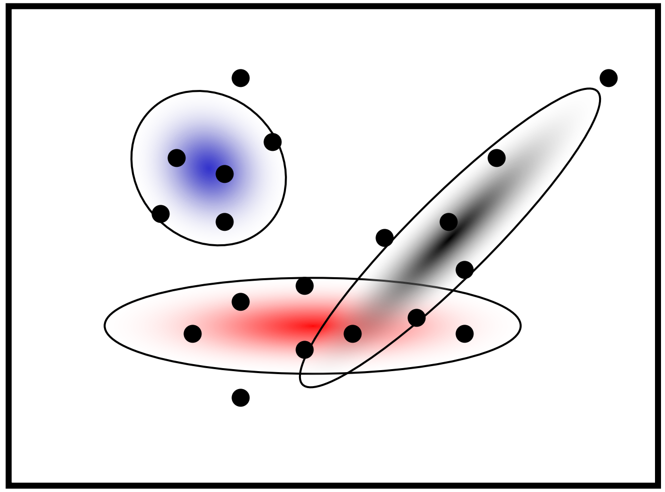
Probabilistic Classification

- Bayesian Classification
  - Given a measurement x, what class C_k should we assign to a pixel?
  - Recall from ML lecture: Bayes Decision Theory
    - We can minimize the probability of misclassification by deciding for the class C_k that has the highest posterior probability

$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{\sum_j p(x|C_j)p(C_j)}$$

    - where
      $p(x|C_k)$ is the likelihood of the measurement x having been generated by class C_k,
      $p(C_k)$ is the prior probability of class C_k.
  - In order to build a classifier, we can either
    - Try to directly estimation the posterior (discriminative methods)
    - Estimate likelihood & prior and then use the above formula (generative methods)

Mixture of Gaussians (MoG)



Each Gaussian blob is defined by its parameters $\theta_j = (\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$

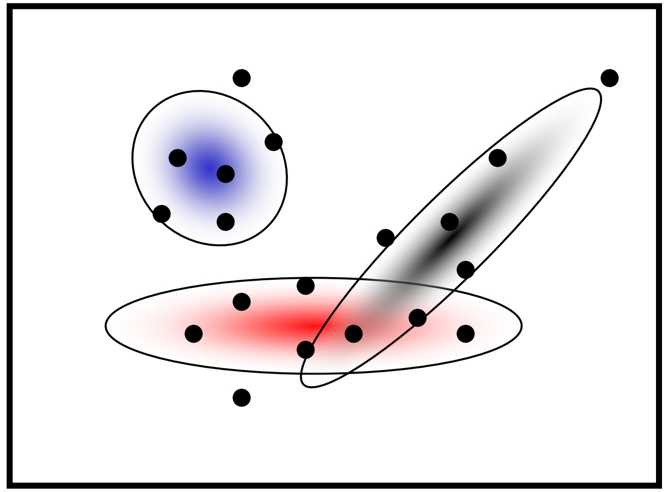
- Generative model for representing probability densities
  - K Gaussian blobs with means $\boldsymbol{\mu}_j$, cov. matrices $\boldsymbol{\Sigma}_j$, dim. D

$$p(\mathbf{x}|\theta_j) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}_j|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_j)^{\mathrm{T}}\boldsymbol{\Sigma}_j^{-1}(\mathbf{x}-\boldsymbol{\mu}_j)\right\}$$

  - Blob j is selected with probability π_j
  - The likelihood of observing $\mathbf{x}$ is a weighted mixture of Gaussians

$$p(\mathbf{x}|\theta) = \sum_{j=1}^{K} \pi_j p(\mathbf{x}|\theta_j) \qquad \theta = (\pi_1, \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1, \ldots, \pi_M, \boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M)$$

Expectation Maximization (EM)



- Goal
  - Find blob parameters θ that maximize the likelihood function:

$$p(data|\theta) = \prod_{n=1}^{N} p(\mathbf{x}_n|\theta)$$

- Approach:
  1. E-step: given current guess of blobs, compute ownership of each point
  2. M-step: given ownership probabilities, update blobs to maximize likelihood function
  3. Repeat until convergence

see lecture Machine Learning!

EM Algorithm

- Expectation-Maximization (EM) Algorithm
  - E-Step: softly assign samples to mixture components

$$\gamma_j(\mathbf{x}_n) \leftarrow \frac{\pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}{\sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)} \qquad \forall j = 1, \ldots, K, \quad n = 1, \ldots, N$$

  - M-Step: re-estimate the parameters (separately for each mixture component) based on the soft assignments

$$\hat{N}_j \leftarrow \sum_{n=1}^{N} \gamma_j(\mathbf{x}_n) \text{ = soft number of samples labeled } j$$

$$\hat{\pi}_j^{\text{new}} \leftarrow \frac{\hat{N}_j}{N}$$

$$\hat{\boldsymbol{\mu}}_j^{\text{new}} \leftarrow \frac{1}{\hat{N}_j} \sum_{n=1}^{N} \gamma_j(\mathbf{x}_n) \mathbf{x}_n$$

$$\hat{\boldsymbol{\Sigma}}_j^{\text{new}} \leftarrow \frac{1}{\hat{N}_j} \sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_j^{\text{new}})(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_j^{\text{new}})^{\mathrm{T}}$$

Slide adapted from Bernt Schiele

How to Apply MoG Models?

- Semantic Segmentation with MoG models
  - Learn one MoG model per semantic class C_k

$$p(x|C_k) = p(x|\theta_{C_k}) = \sum_{j=1}^{M_k} \pi_{k,j}\, p(x|\theta_{k,j})$$

"per-class likelihood"

$\Rightarrow$ E.*g., learn a class-specific color model*

  - Perform probabilistic classification by evaluating the posterior

$$p(C_k|x) = \frac{p(x|C_k)p(C_k)}{\sum_j p(x|C_j)p(C_j)}$$

"posterior probability"

$\Rightarrow$ Decide for the class with the highest posterior

MoG Color Models for Image Segmentation

(a) input image (b) user input (c) inferred segmentation

- User assisted image segmentation
  - User marks two regions for foreground and background.
  - Learn a MoG model for the color values in each region.
  - Use those models to classify all other pixels by deciding for the class with the highest posterior probability.

  $\Rightarrow$ Simple segmentation procedure
  (building block for more complex applications)

Summary: Mixtures of Gaussians, EM

- Pros
  - Probabilistic interpretation
  - Soft assignments between data points and clusters
  - Generative model, can predict novel data points
  - Relatively compact storage
- Cons
  - Local minima
    - EM will only converge to a local optimum of the likelihood
  - Slow convergence
    - Often a good idea to start with some k-means iterations.
  - Need to know number of components
    - Solutions: Model selection ($\rightarrow$ Machine Learning literature)
  - Needs careful implementation to avoid numerical issues

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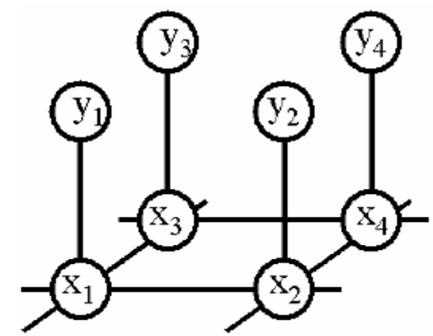
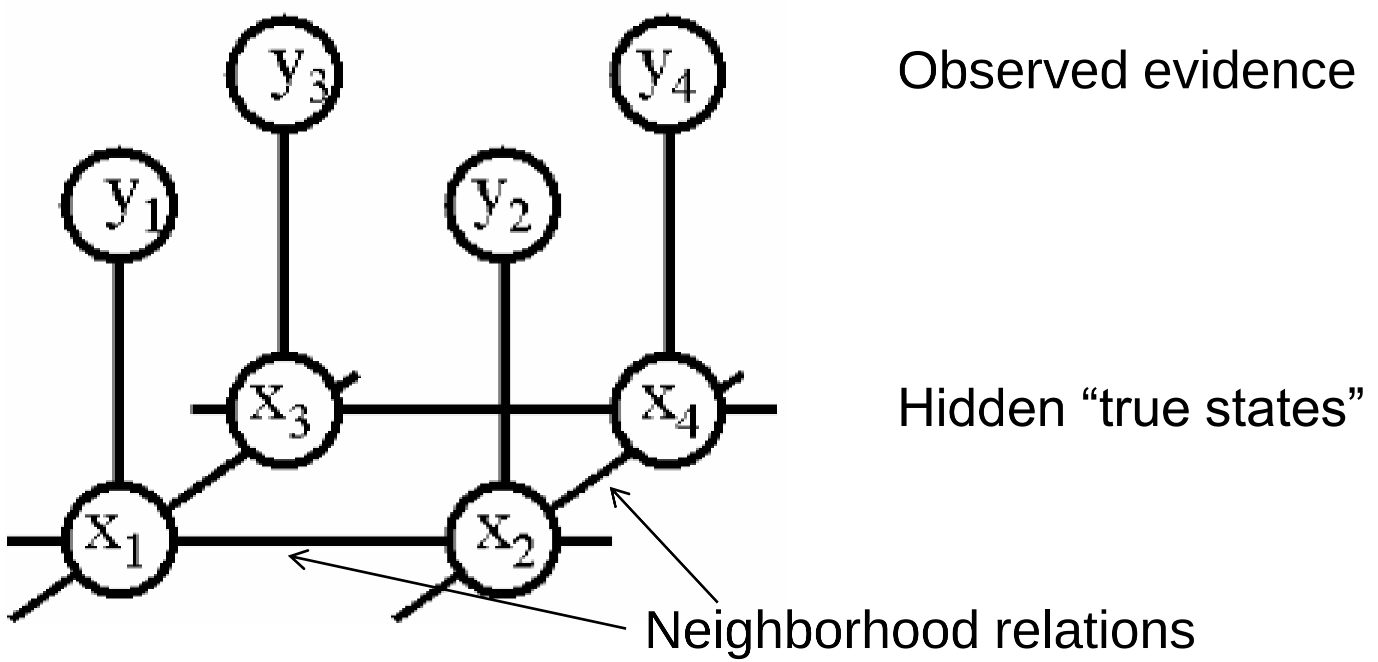


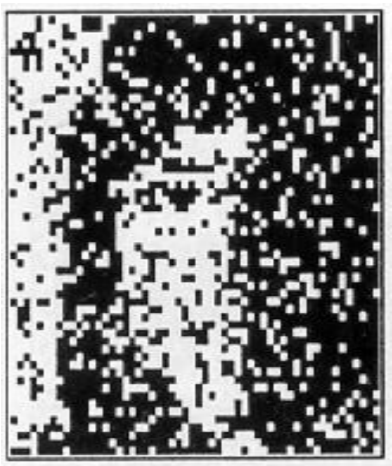
Markov Random Fields

- Allow rich probabilistic models for images
- But built in a local, modular way
  - Learn local effects, get global effects out



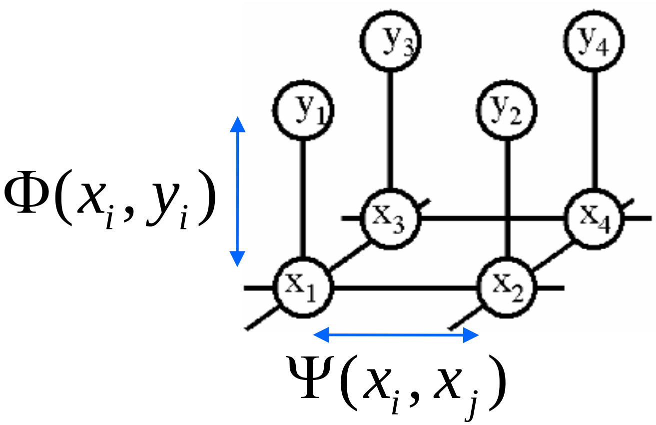
MRF Nodes as Pixels

Original image

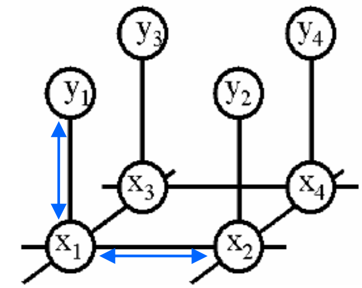


Degraded image

Reconstruction from MRF modeling pixel neighborhood statistics



Network Joint Probability



$$p(\mathbf{x}, \mathbf{y}) = \prod_i \Phi(x_i, y_i) \prod_{i,j} \Psi(\underbrace{x_i, x_j}_{})$$

Scene

Image

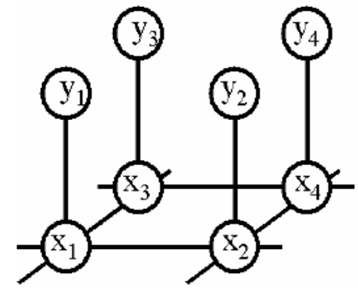
Image-scene compatibility function

Local observations

Scene-scene compatibility function

Neighboring scene nodes

Energy Formulation



- Joint probability

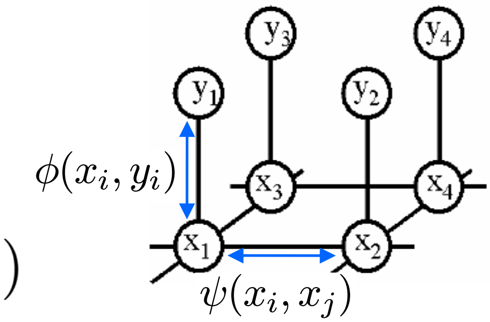
$$p(\mathbf{x},\mathbf{y}) = \prod_i \Phi(x_i, y_i) \prod_{i,j} \Psi(x_i, x_j)$$

- Maximizing the joint probability is the same as minimizing the negative logarithm of it

$$\begin{aligned} -\log p(\mathbf{x},\mathbf{y}) &= -\sum_i \log \Phi(x_i, y_i) - \sum_{i,j} \log \Psi(x_i, x_j) \\ E(\mathbf{x},\mathbf{y}) &= \sum_i \phi(x_i, y_i) + \sum_{i,j} \psi(x_i, x_j) \end{aligned}$$

- This is similar to free-energy problems in statistical mechanics (spin glass theory). We therefore draw the analogy and call E an energy function.
- ϕ and ψ are called potentials.

Energy Formulation

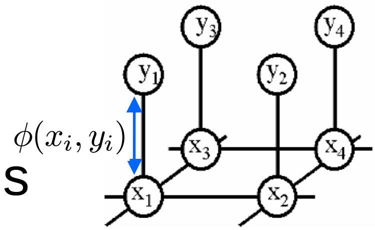


- Energy function

$$E(\mathbf{x},\mathbf{y}) = \sum_i \underbrace{\phi(x_i,y_i)}_{\text{Single-node potentials}} + \sum_{i,j} \underbrace{\psi(x_i,x_j)}_{\text{Pairwise potentials}}$$

- Single-node potentials ϕ ("unary potentials")
  - Encode local information about the given pixel/patch
  - How likely is a pixel/patch to belong to a certain class (e.g., foreground/background)?
- Pairwise potentials ψ
  - Encode neighborhood information
  - How different is a pixel/patch's label from that of its neighbor? (e.g., based on intensity/color/texture difference, edges)

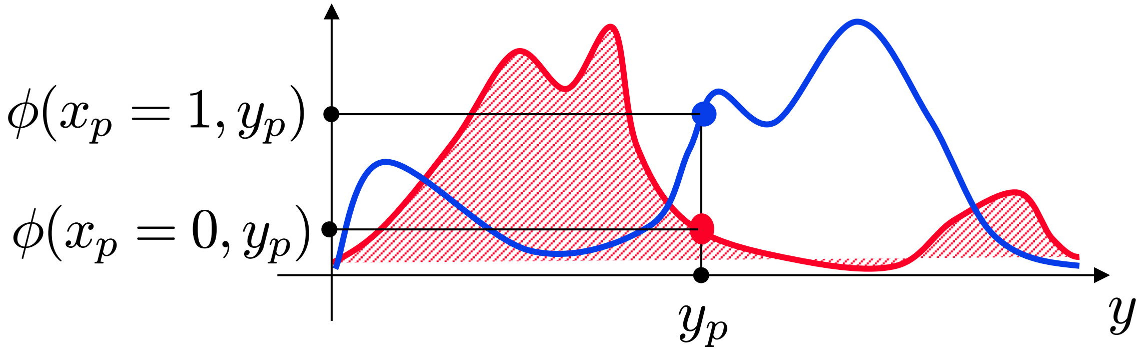
How to Set the Potentials? Some Examples



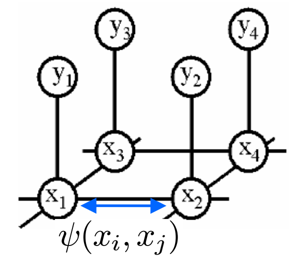
- Unary potentials
  - E.g., color model, modeled with a Mixture of Gaussians

$$\phi(x_i, y_i) = -\theta_\phi \log \sum_k \pi_{k,x_i} \mathcal{N}(y_i; \boldsymbol{\mu}_{k,x_i}, \boldsymbol{\Sigma}_{k,x_i})$$

$\Rightarrow$ Learn color distributions for each label



How to Set the Potentials? Some Examples



- Pairwise potentials
  - Simplest choice: "Potts model"

$$\psi(x_i, x_j) = \theta_\psi \, \delta(x_i \neq x_j)$$

    – Effect: Fixed penalty for all label changes

  - Extension: "contrast-sensitive Potts model"

$$\psi(x_i, x_j, g_{ij}(\mathbf{y})) = \theta_\psi \, g_{ij}(\mathbf{y}) \, \delta(x_i \neq x_j)$$

where

$$g_{ij}(\mathbf{y}) = e^{-\beta ||y_i - y_j||^2} \qquad \beta = 2 \operatorname{avg}\left(||y_i - y_j||^2\right)$$

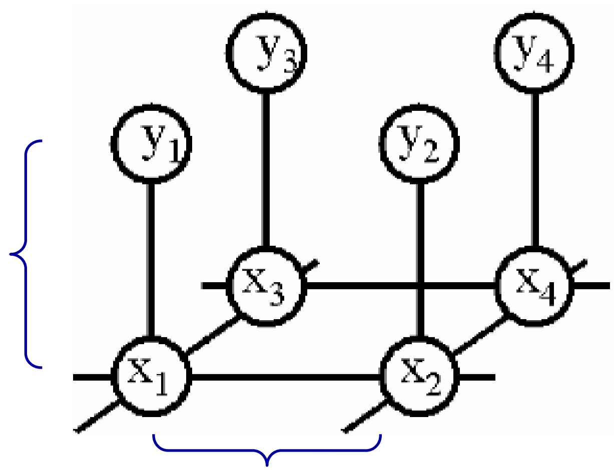
    – Effect: Discourages label changes except in places where there is also a large change in the observations.

[Shotton & Winn, ECCV'06]

Example: MRF for Image Segmentation

- MRF structure

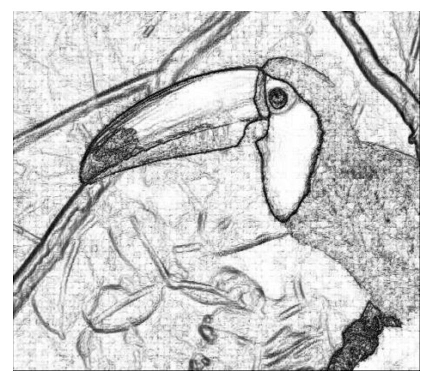
unary potentials



pairwise potentials

Data (D)

Unary likelihood

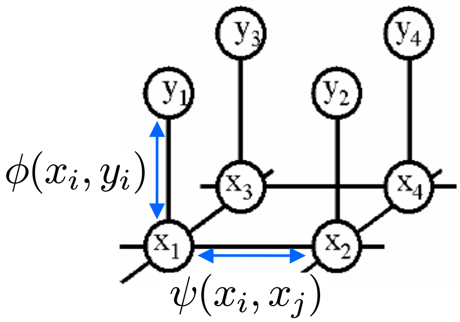


Pair-wise Terms

MAP Solution

Slide adapted from Phil Torr

Energy Minimization



- Goal:
  - Infer the optimal labeling of the MRF.
- Many inference algorithms are available, e.g.
  - Gibbs sampling, simulated annealing
  - Iterated conditional modes (ICM)
  - Variational methods
  - Belief propagation
  - Graph cuts

see lecture Advanced ML!

- Graph Cuts have become a popular tool
  - Only suitable for a certain class of energy functions
  - But the solution can be obtained very fast for typical vision problems (>1MPixel/sec).

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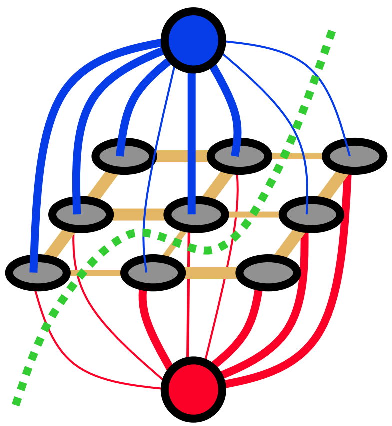
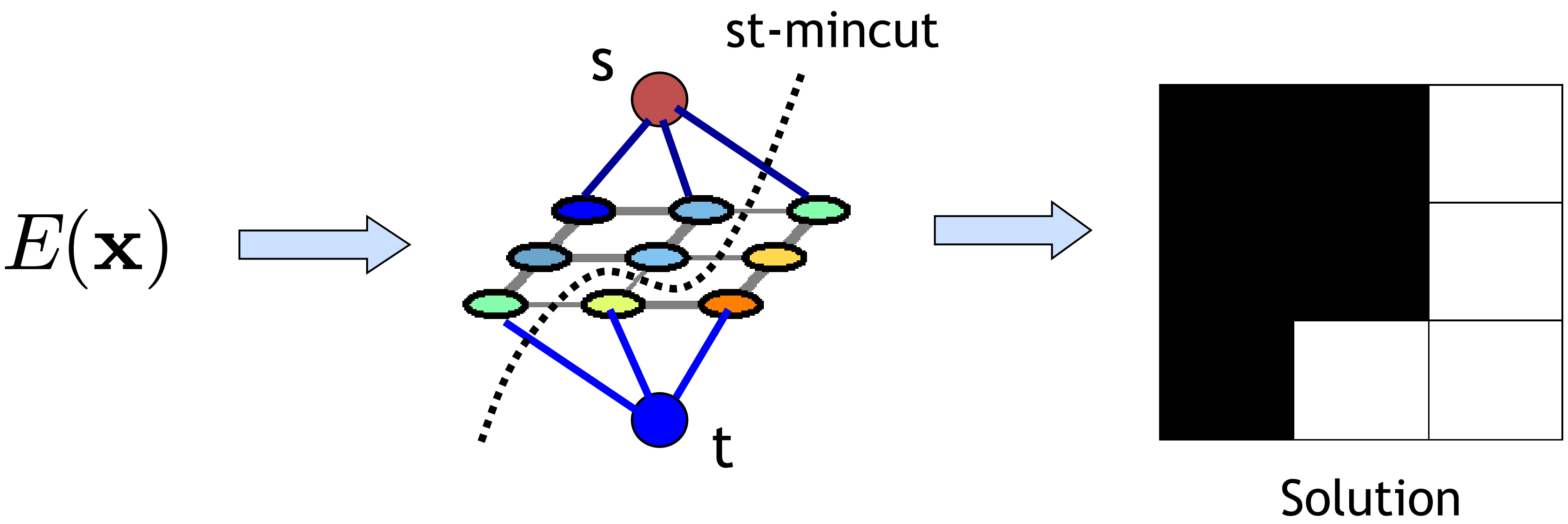


Graph Cuts – Basic Idea

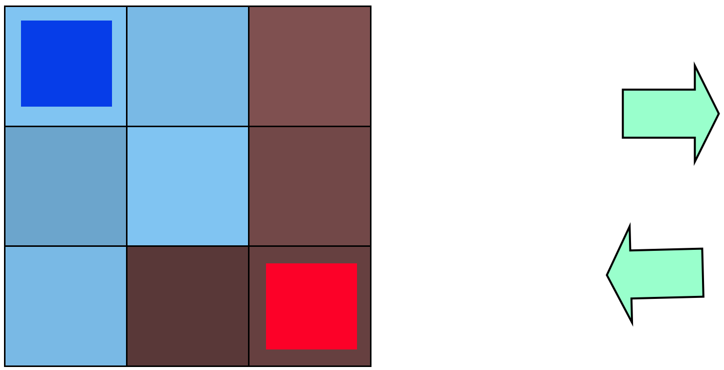
- Construct a graph such that:
  1. Any st-cut corresponds to an assignment of $\mathbf{x}$
  2. The cost of the cut is equal to the energy of $\mathbf{x}$: $E(\mathbf{x})$

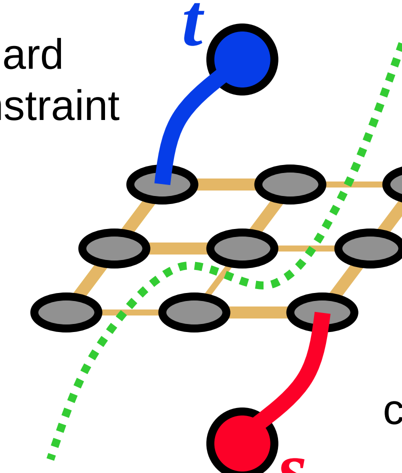


Slide credit: Pushmeet Kohli

Graph Cuts for Optimal Boundary Detection

- Idea: convert MRF into a source-sink graph





Minimum cost cut can be computed in polynomial time

(max-flow/min-cut algorithms)

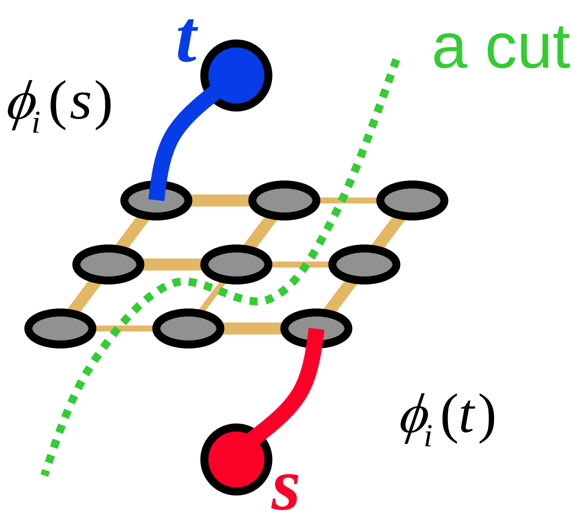
[Boykov & Jolly, ICCV'01]

Simple Example of Energy

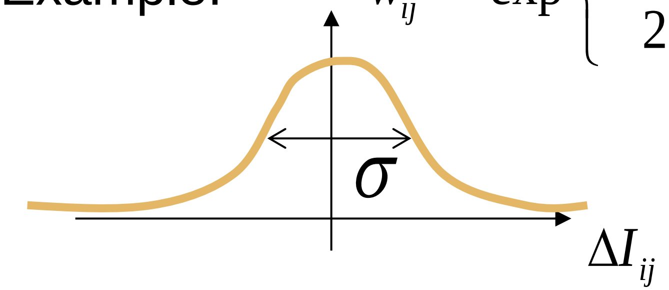
$$E(\mathbf{x},\mathbf{y}) = \sum_i \phi_i(x_i) + \sum_{i,j} w_{ij} \cdot \delta(x_i \neq x_j)$$

Unary terms

Pairwise terms ("Potts model")



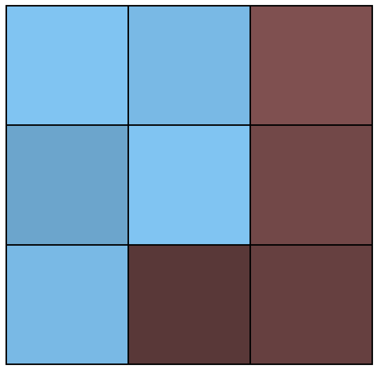
Example: $w_{ij} = \exp\left\{-\frac{\Delta I_{ij}}{2\sigma^2}\right\}$

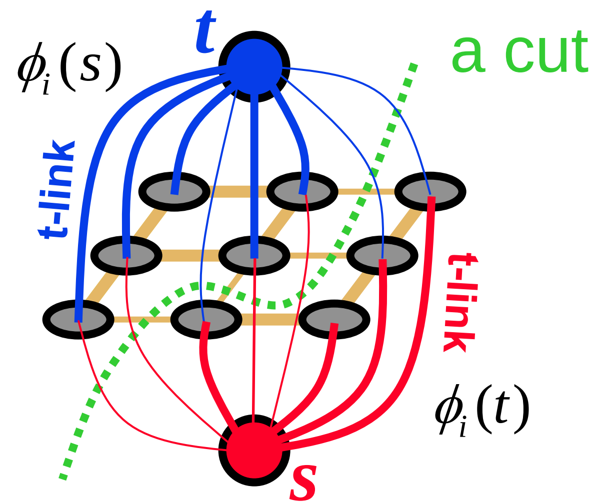


$$x \in \{s,t\}$$

(binary object segmentation)

Adding Regional Properties (Example)





Suppose I^s and I^t are given "expected" intensities of object and background

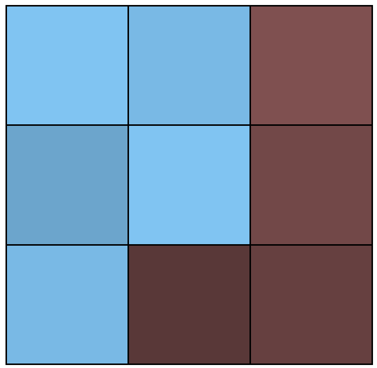
$$\phi_i(s) \propto \exp\left(-\| I_i - I^s \|^2 / 2\sigma^2\right)$$

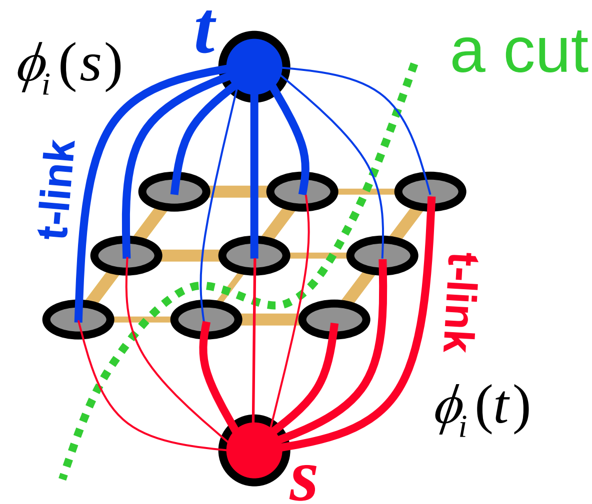
$$\phi_i(t) \propto \exp\left(-\| I_i - I^t \|^2 / 2\sigma^2\right)$$

NOTE: hard constrains are not required, in general.

[Boykov & Jolly, ICCV'01]

Adding Regional Properties (Example)





"expected" intensities of
object and background
I^s and I^t
can be re-estimated

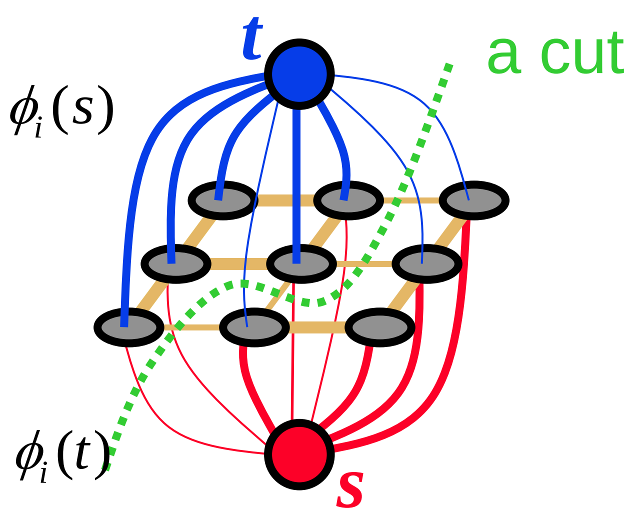
$$\phi_i(s) \propto \exp\left(-\| I_i - I^s \|^2 / 2\sigma^2\right)$$

$$\phi_i(t) \propto \exp\left(-\| I_i - I^t \|^2 / 2\sigma^2\right)$$

EM-style optimization

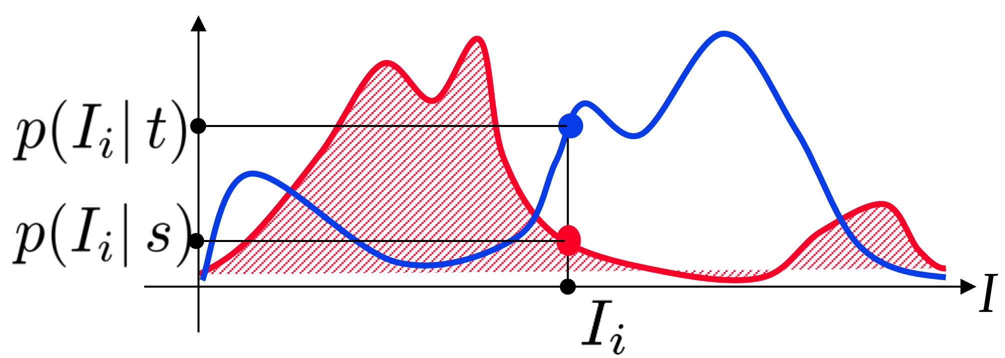
Adding Regional Properties

- More generally, unary potentials can be based on any intensity/color models of object and background



Note: $\phi(t)$ is the cost for the link to the s node!
Why?

$$\phi_i(L_i) = -\log p(I_i|L_i)$$

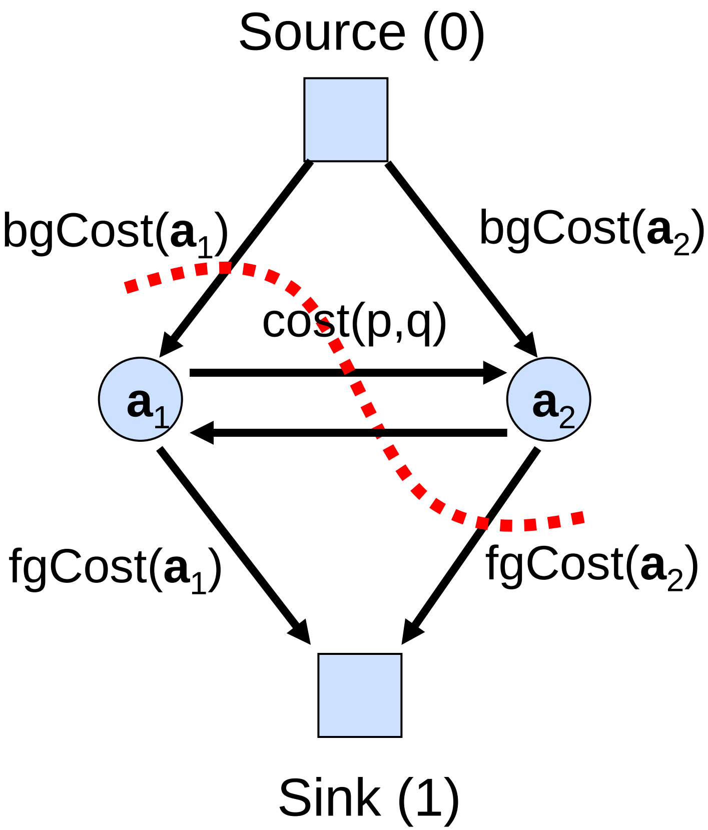


given object and background intensity histograms

[Boykov & Jolly, ICCV'01]

How Does the Code Look Like?

```
Graph *g;

For all pixels p

        /* Add a node to the graph */
        nodeID(p) = g->add_node();

        /* Set cost of terminal edges */
        set_weights(nodeID(p), fgCost(p), bgCost(p));

end

for all adjacent pixels p,q
        add_weights(nodeID(p), nodeID(q),  cost(p,q));
end

g->compute_maxflow();

label_p = g->is_connected_to_source(nodeID(p));

// is the label of pixel p (0 or 1)
```



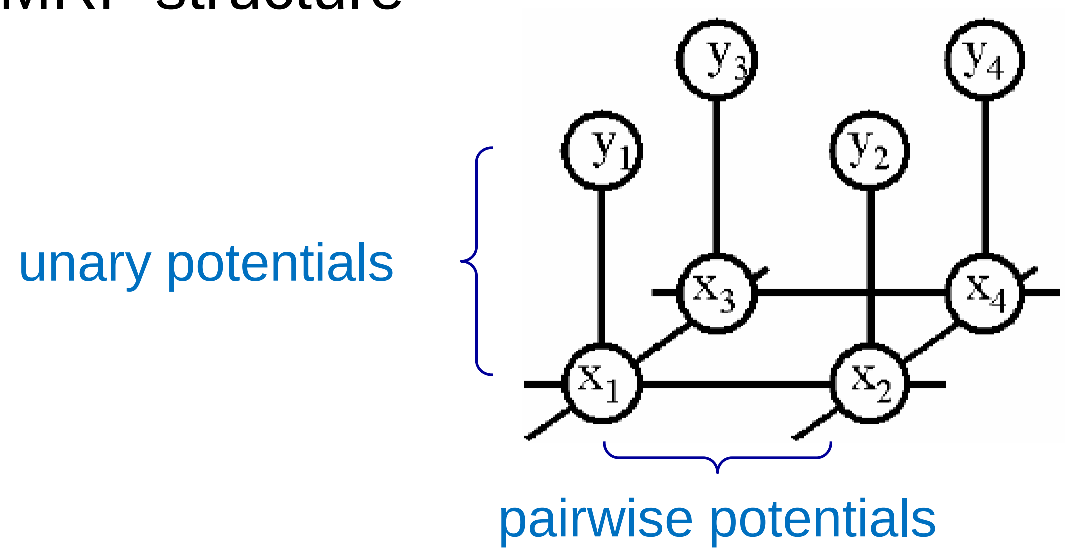
$\boxed{\mathbf{a}_1 = \text{bg} \quad \mathbf{a}_2 = \text{fg}}$

Slide credit: Carsten Rother

Computer Vision

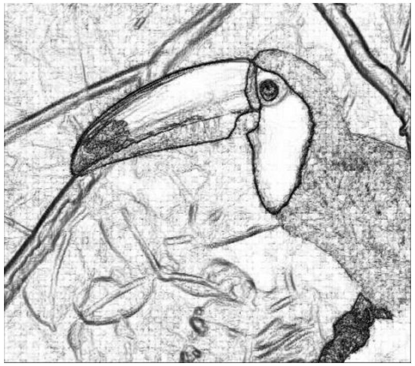
Example: MRF for Image Segmentation

- MRF structure



Data (D)

Unary likelihood



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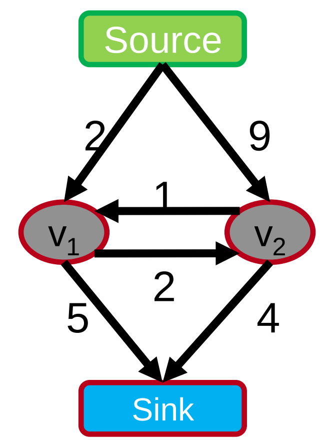
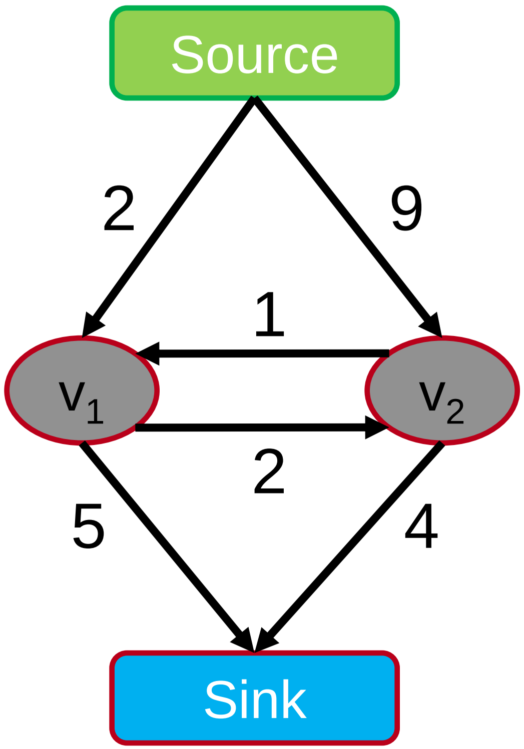


How Does it Work? The s-t-Mincut Problem



Graph (V, E, C)

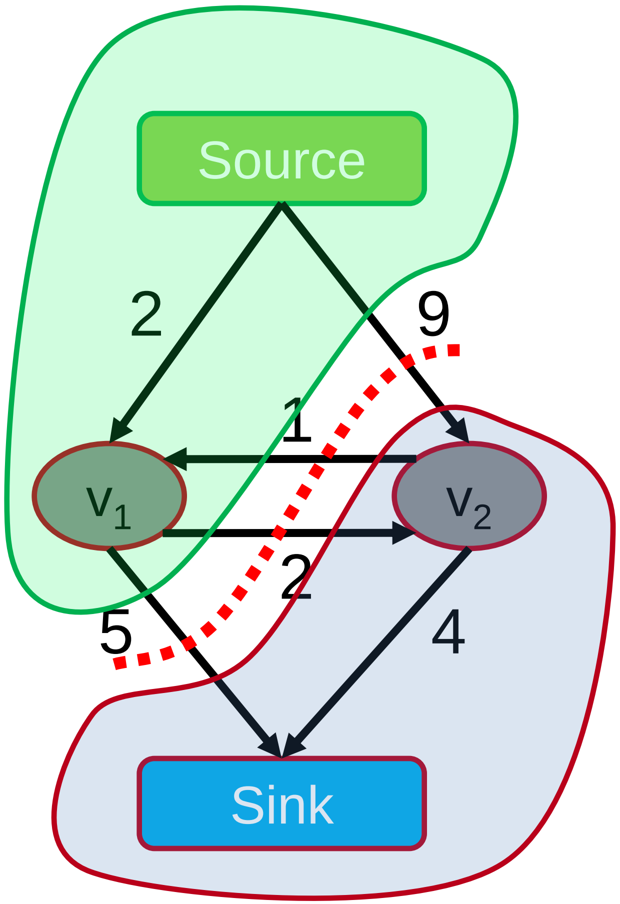
Vertices V = $\{v_1, v_2 \ldots v_n\}$

Edges E = $\{(v_1, v_2) \ldots .\}$

Costs C = $\{c_{(1, 2)} \ldots .\}$

Slide credit: Pushmeet Kohli

The s-t-Mincut Problem



5 + 2 + 9 = 16

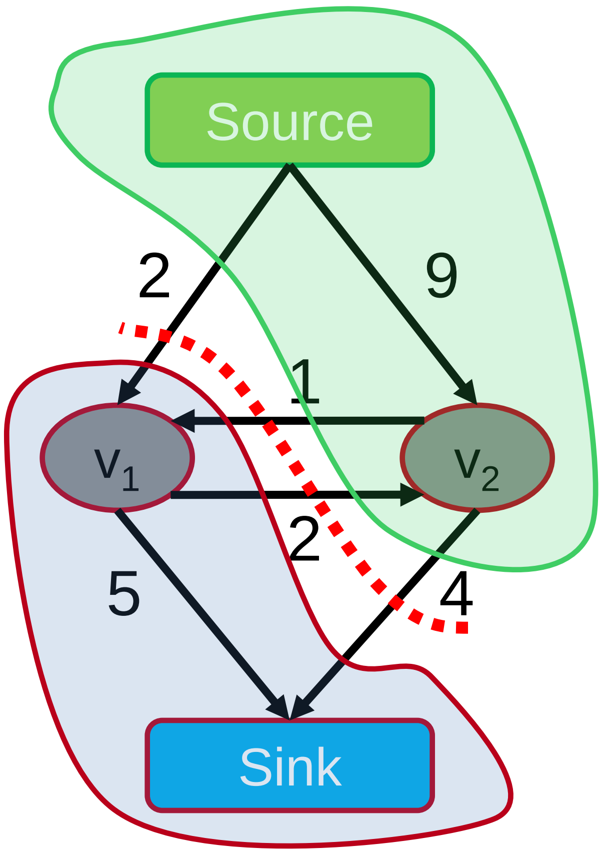
What is an st-cut?

An st-cut (S,T) divides the nodes between source and sink.

What is the cost of a st-cut?

Sum of cost of all edges going from S to T

The s-t-Mincut Problem



$2 + 1 + 4 = 7$

What is an st-cut?

An st-cut (S,T) divides the nodes between source and sink.

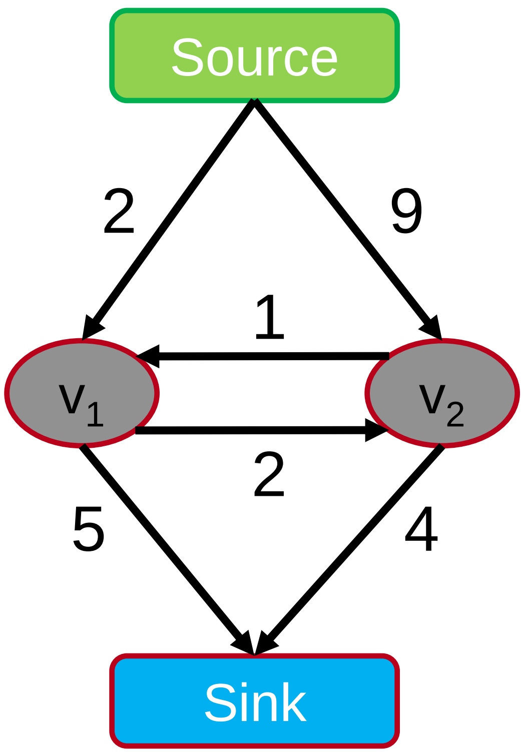
What is the cost of a st-cut?

Sum of cost of all edges going from S to T

What is the st-mincut?

st-cut with the minimum cost

How to Compute the s-t-Mincut?



Solve the dual maximum flow problem

Compute the maximum flow between Source and Sink

Constraints

Edges: Flow < Capacity

Nodes: Flow in = Flow out

Min-cut/Max-flow Theorem

In every network, the maximum flow equals the cost of the st-mincut

History of Maxflow Algorithms

Augmenting Path and Push-Relabel

year	discoverer(s)	bound
1951	Dantzig	$O(n^2mU)$
1955	Ford & Fulkerson	$O(m^2U)$
1970	Dinitz	$O(n^2m)$
1972	Edmonds & Karp	$O(m^2 \log U)$
1973	Dinitz	$O(nm \log U)$
1974	Karzanov	$O(n^3)$
1977	Cherkassky	$O(n^2m^{1/2})$
1980	Galil & Naamad	$O(nm \log^2 n)$
1983	Sleator & Tarjan	$O(nm \log n)$
1986	Goldberg & Tarjan	$O(nm \log(n^2/m))$
1987	Ahuja & Orlin	$O(nm + n^2 \log U)$
1987	Ahuja et al.	$O(nm \log(n\sqrt{\log U}/m))$
1989	Cheriyan & Hagerup	$E(nm + n^2 \log^2 n)$
1990	Cheriyan et al.	$O(n^3/\log n)$
1990	Alon	$O(nm + n^{8/3} \log n)$
1992	King et al.	$O(nm + n^{2+\epsilon})$
1993	Phillips & Westbrook	$O(nm(\log_{m/n} n + \log^{2+\epsilon} n))$
1994	King et al.	$O(nm \log_{m/(n \log n)} n)$
1997	Goldberg & Rao	$O(m^{3/2} \log(n^2/m) \log U)$ $O(n^{2/3}m \log(n^2/m) \log U)$

n: #nodes

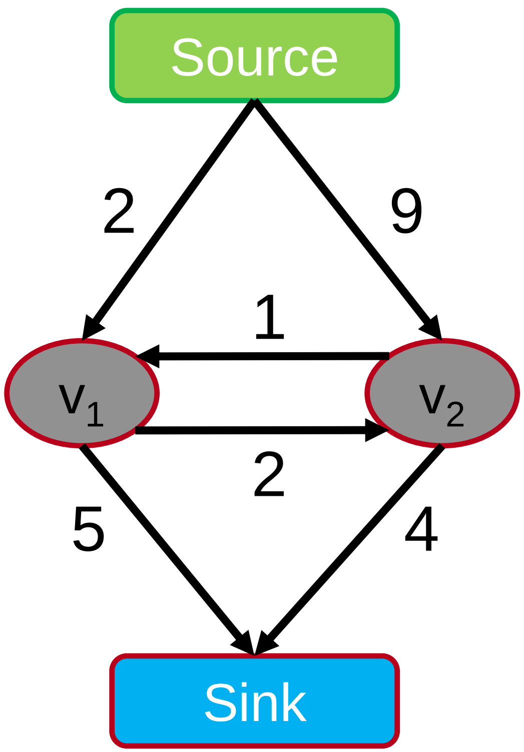
m: #edges

U: maximum edge weight

Algorithms assume non-negative edge weights

Maxflow Algorithms

Flow = 0



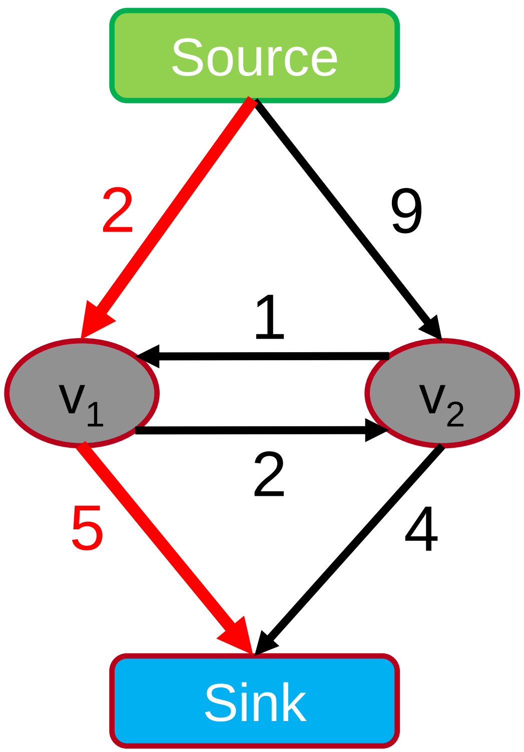
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 0



Augmenting Path Based Algorithms

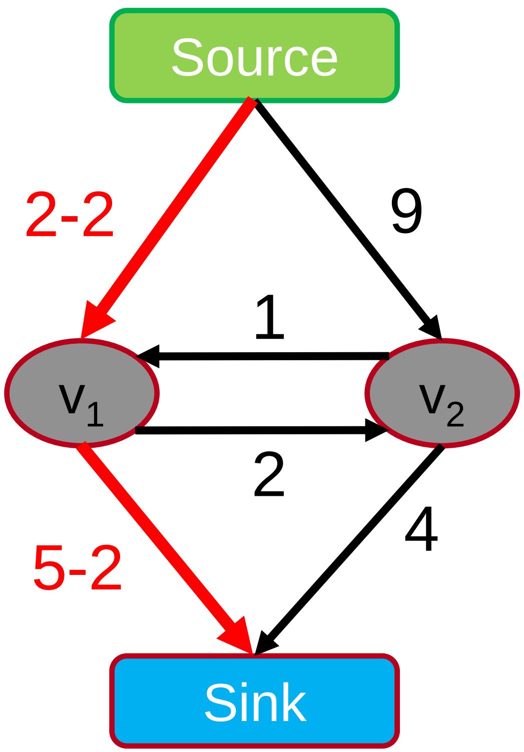
1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Slide credit: Pushmeet Kohli

Maxflow Algorithms

Flow = 0 + 2



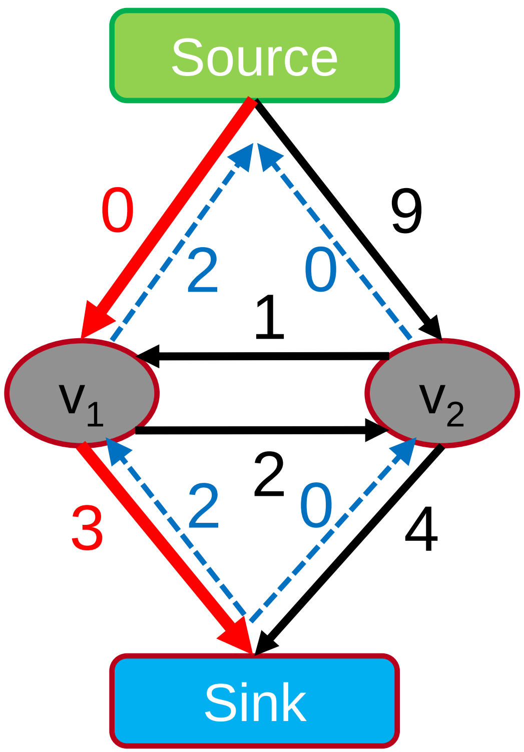
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 2



“Residual flows”

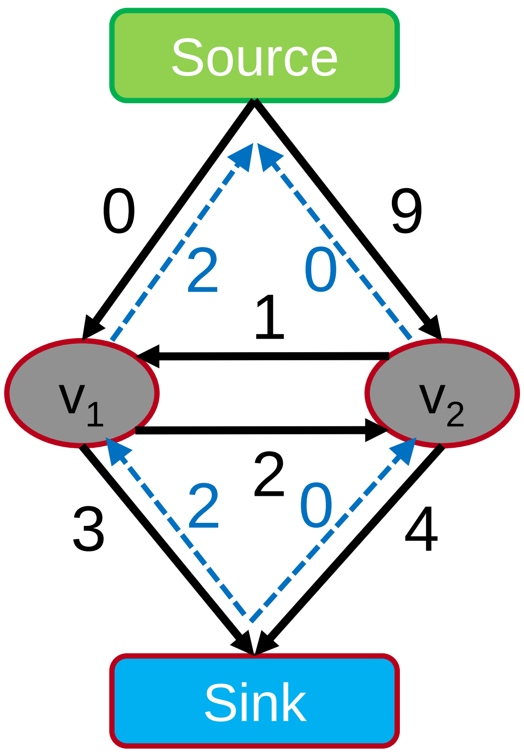
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges and record “residual flows”
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 2

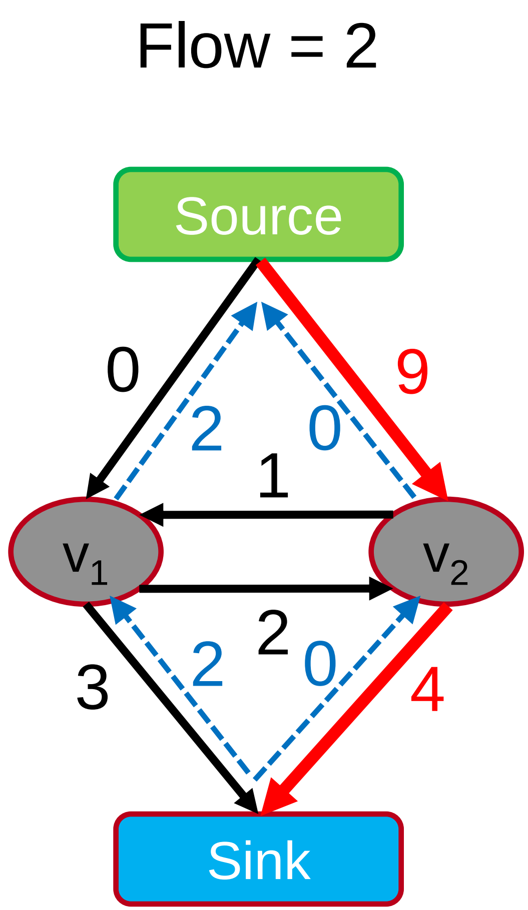


Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms



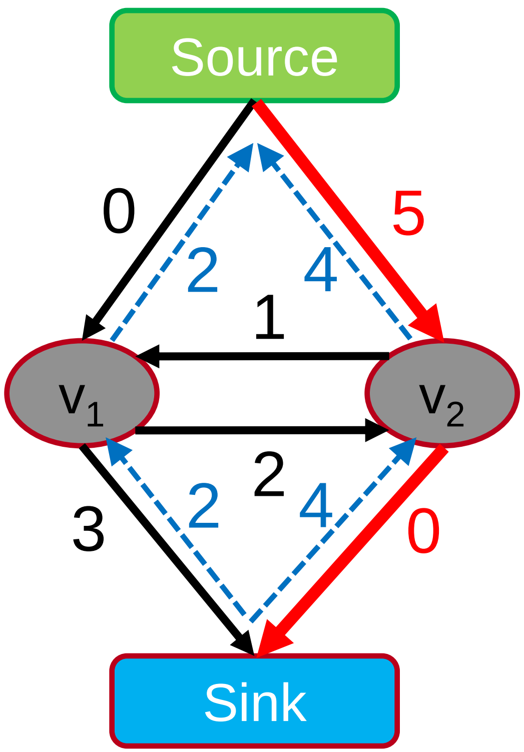
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 2 + 4



Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
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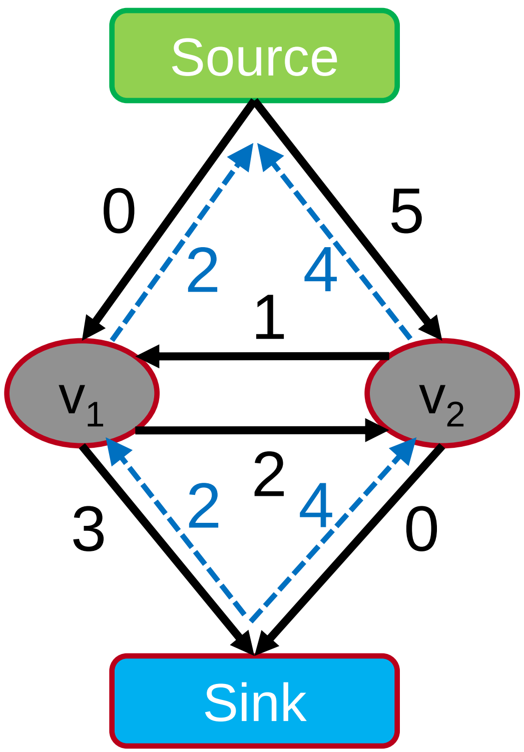
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 6



Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
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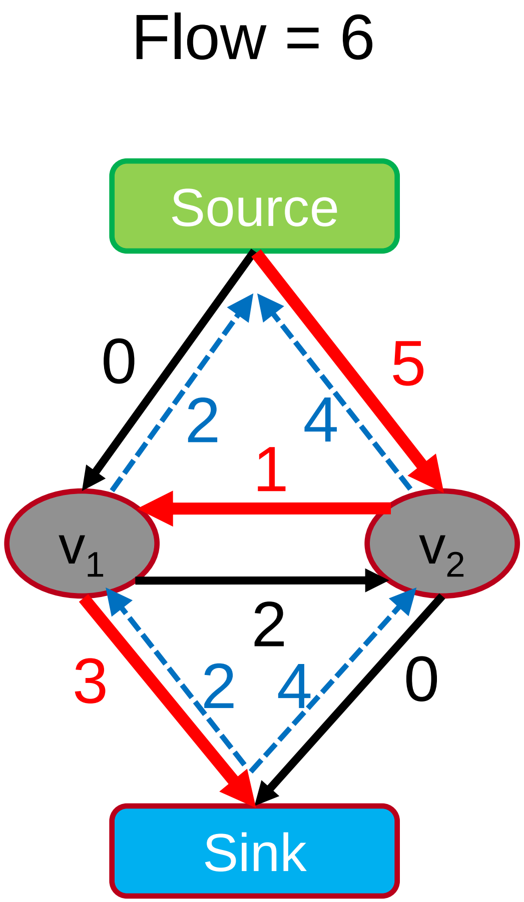
Algorithms assume non-negative capacity

Slide credit: Pushmeet Kohli

Maxflow Algorithms



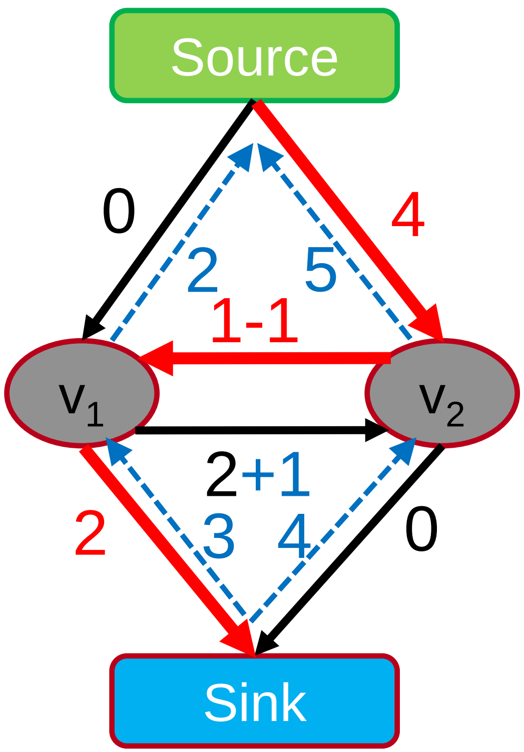
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 6 + 1



Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
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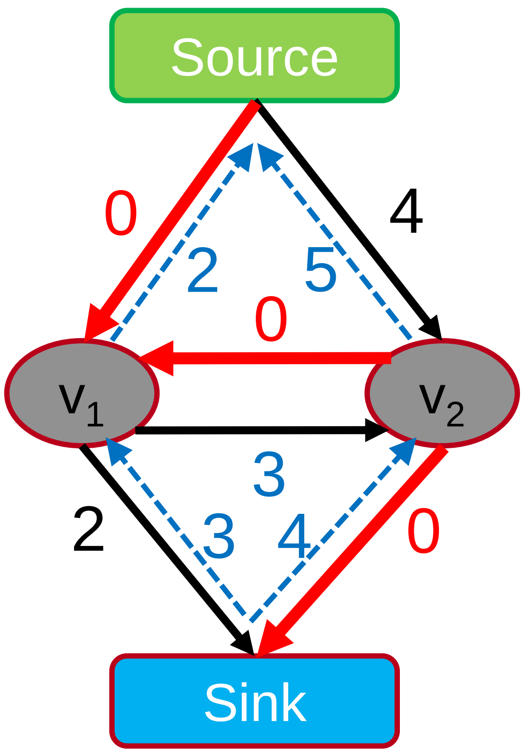
4. Repeat until no path can be found

Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 7



Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
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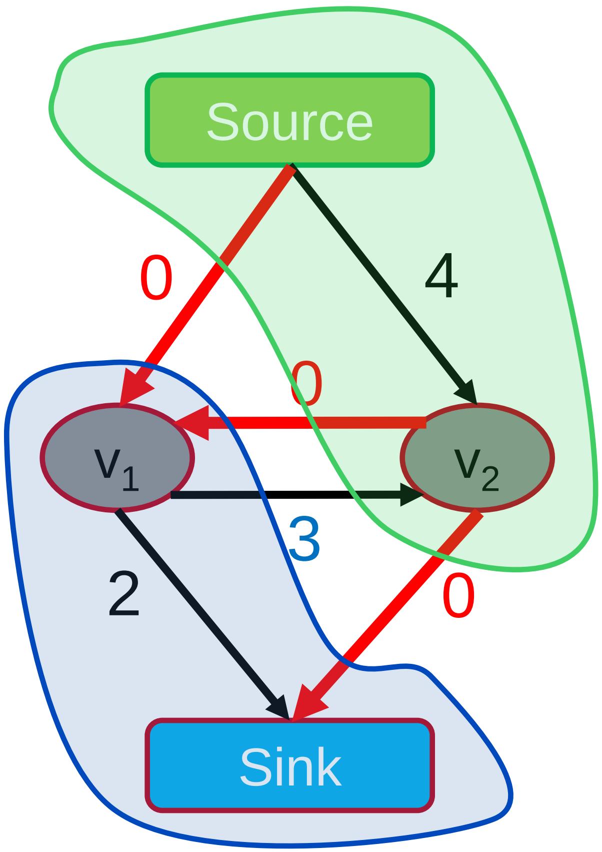
Algorithms assume non-negative capacity

Maxflow Algorithms

Flow = 7



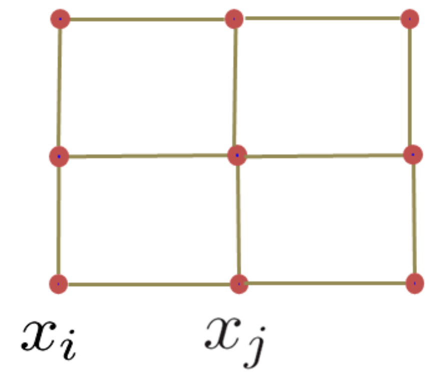
Augmenting Path Based Algorithms

1. Find path from source to sink with positive capacity
2. Push maximum possible flow through this path
3. Adjust the capacity of the used edges
4. Repeat until no path can be found

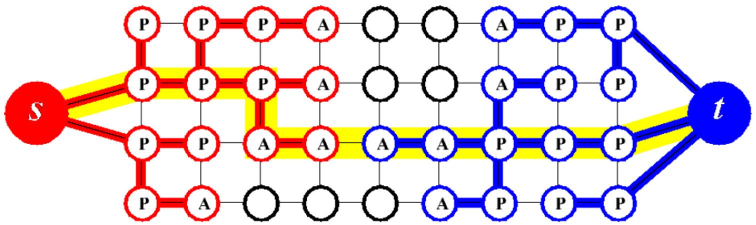
Algorithms assume non-negative capacity

Applications: Maxflow in Computer Vision

- Specialized algorithms for vision problems
  - Grid graphs
  - Low connectivity (m ~ O(n))



- Dual search tree augmenting path algorithm
  [Boykov and Kolmogorov PAMI 2004]
  - Finds approximate shortest augmenting paths efficiently.
  - High worst-case time complexity.
  - Empirically outperforms other algorithms on vision problems.
  - Efficient code available on the web
    http://pub.ist.ac.at/~vnk/software.html



When Can s-t Graph Cuts Be Applied?

$$E(L) = \underbrace{\sum_{p} E_p(L_p)}_{\text{t-links}}^{\text{Unary potentials}} + \underbrace{\sum_{pq \in N} E(L_p, L_q)}_{\text{n-links}}^{\text{Pairwise potentials}} \qquad L_p \in \{s,t\}$$

- s-t graph cuts can only globally minimize binary energies that are submodular. [Boros & Hummer, 2002, Kolmogorov & Zabih, 2004]

$E(L)$ can be minimized by *s-t* graph cuts $\Leftrightarrow$ $\boxed{E(s,s) + E(t,t) \leq E(s,t) + E(t,s)}$

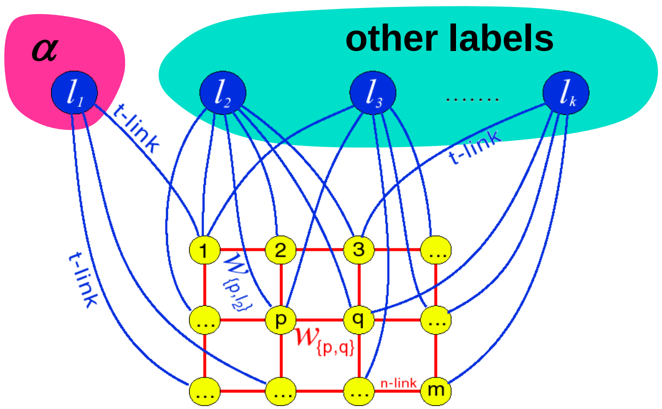
Submodularity ("convexity")

- Submodularity is the discrete equivalent to convexity.
  - Implies that every local energy minimum is a global minimum.

  $\Rightarrow$ Solution will be globally optimal.

Topics of This Lecture

- Segmentation as pixel-wise classification
  - Probabilistic classification
  - Mixture of Gaussians, EM
- Segmentation as energy minimization
  - Markov Random Fields
  - Energy formulation
- Graph cuts for image segmentation
  - Basic idea
  - s-t Mincut algorithm
  - Extension to non-binary case

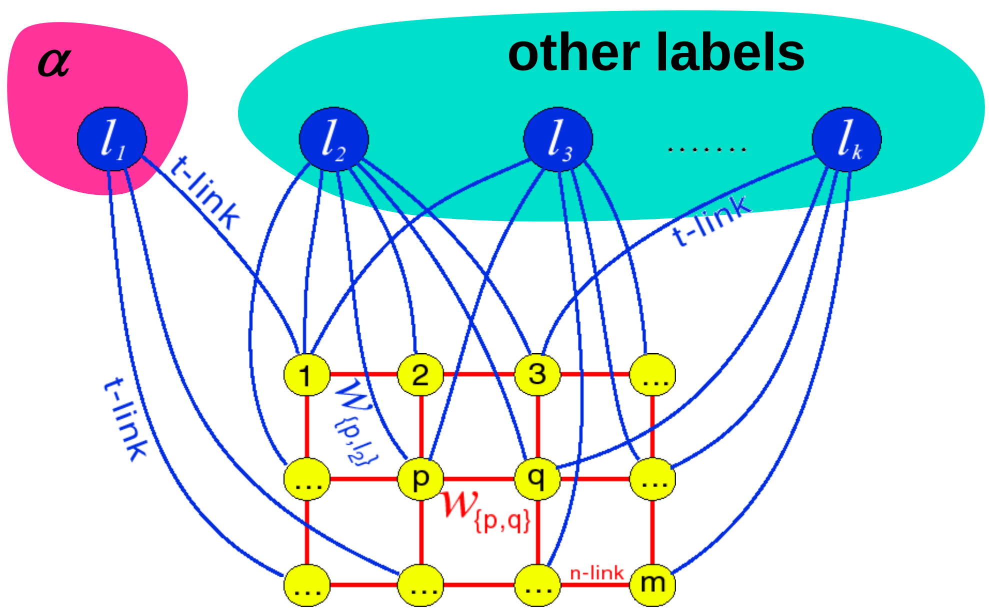


Dealing with Non-Binary Cases

- Limitation to binary energies is often a nuisance.
  $\Rightarrow$ E.g. binary segmentation only…
- We would like to solve also multi-label problems.
  - The bad news: Problem is NP-hard with 3 or more labels!
- There exist some approximation algorithms which extend graph cuts to the multi-label case:
  - α-Expansion
  - $\alpha\beta$-Swap
- They are no longer guaranteed to return the globally optimal result.
  - But α-Expansion has a guaranteed approximation quality (2-approx) and converges in a few iterations.

α-Expansion Move

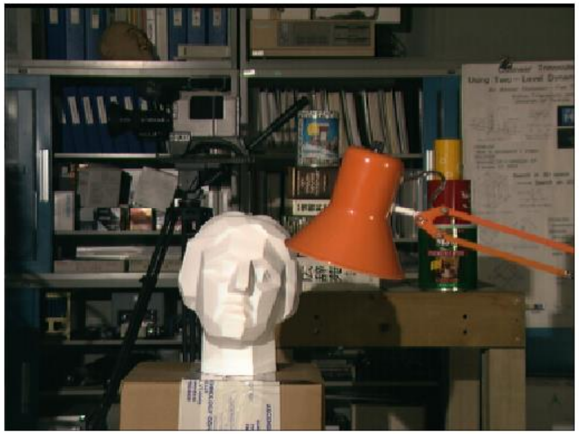
- Basic idea:
  - Break multi-way cut computation into a sequence of binary s-t cuts.

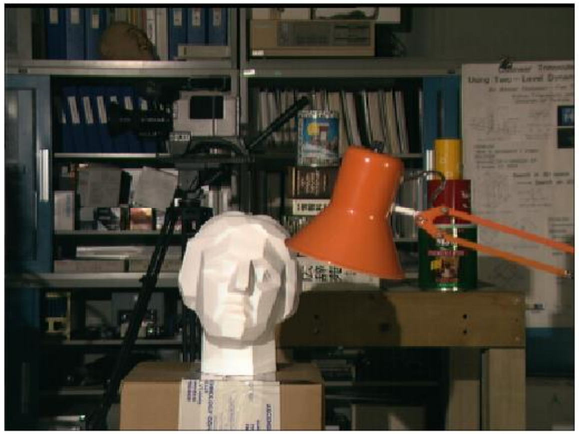


α-Expansion Algorithm

1. Start with any initial solution
2. For each label "α" in any (e.g. random) order:
   1. Compute optimal α-expansion move (s-t graph cuts).
   2. Decline the move if there is no energy decrease.
3. Stop when no expansion move would decrease energy.

Example: Stereo Vision





Original pair of “stereo” images

Depth map

α-Expansion Moves

- In each α-expansion a given label "α" grabs space from other labels

initial solution

-expansion

-expansion

-expansion

-expansion

-expansion

-expansion

-expansion

For each move, we choose the expansion that gives the largest decrease in the energy: $\Rightarrow$ binary optimization problem

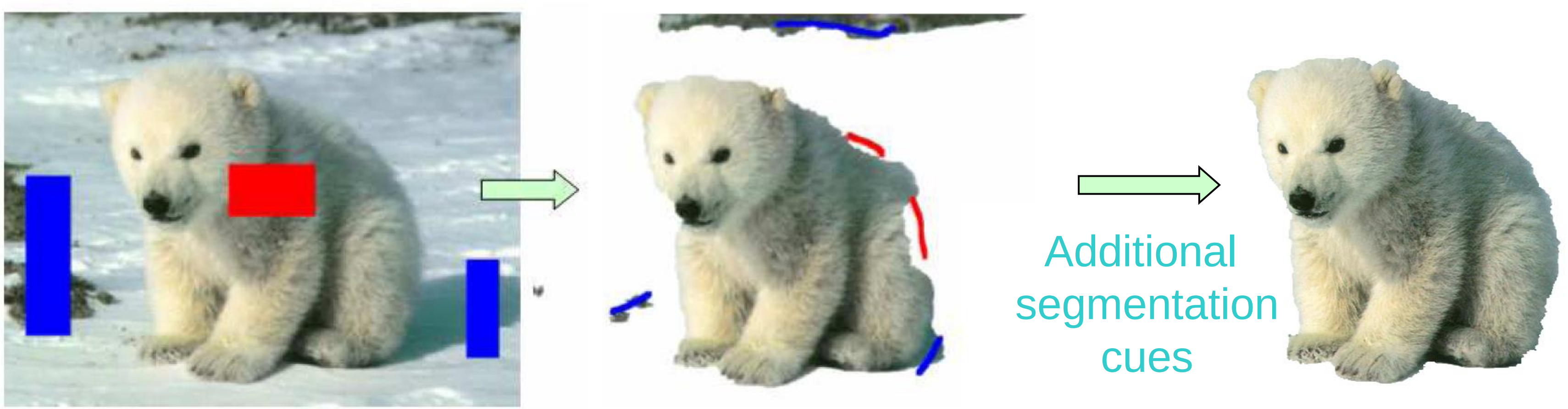
Slide credit: Yuri Boykov

Topics of This Lecture

- Segmentation as pixel-wise classification
  - Probabilistic classification
  - Mixture of Gaussians, EM
- Segmentation as energy minimization
  - Markov Random Fields
  - Energy formulation
- Graph cuts for image segmentation
  - Basic idea
  - s-t Mincut algorithm
  - Extension to non-binary case
- Applications
  - Interactive segmentation

GraphCut Applications: "GrabCut"

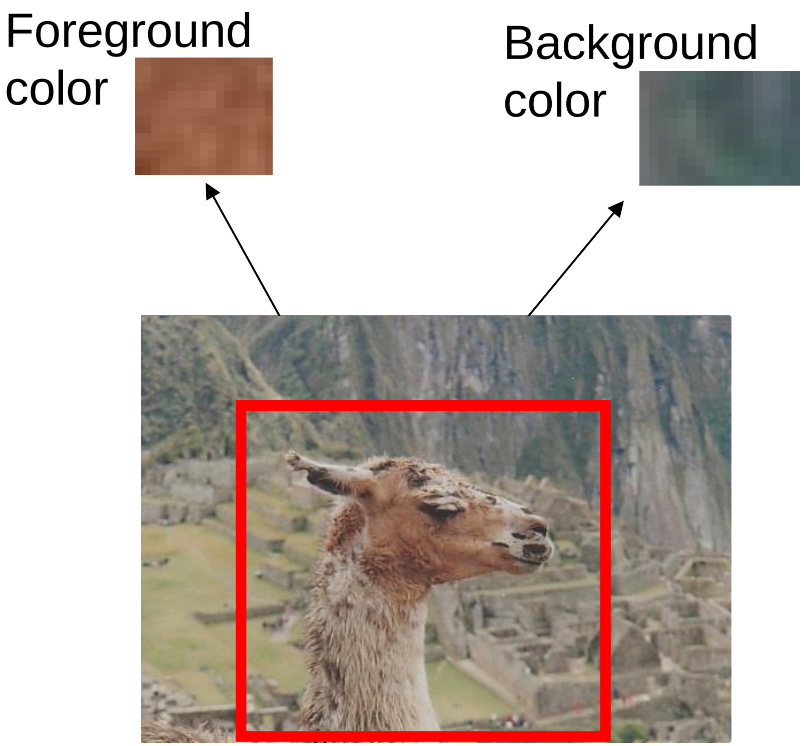
- Interactive Image Segmentation [Boykov & Jolly, ICCV'01]
  - Rough region cues sufficient
  - Segmentation boundary can be extracted from edges
- Procedure
  - User marks foreground and background regions with a brush.
  - This is used to create an initial segmentation which can then be corrected by additional brush strokes.



User segmentation cues

Slide credit: Matthieu Bray

GrabCut: Data Model



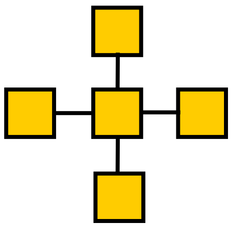
Global optimum of the energy

- Obtained from interactive user input
  - User marks foreground and background regions with a brush
  - Alternatively, user can specify a bounding box

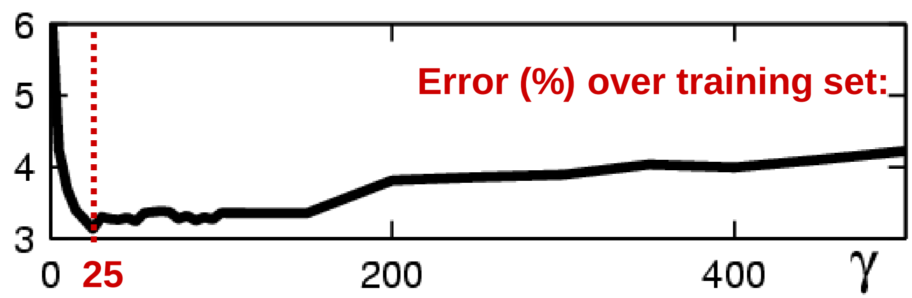
GrabCut: Coherence Model

- An object is a coherent set of pixels:

$$\psi(x,y) = \gamma \sum_{(m,n)\in C} \delta[x_n \neq x_m] e^{-\beta \|y_m - y_n\|^2}$$

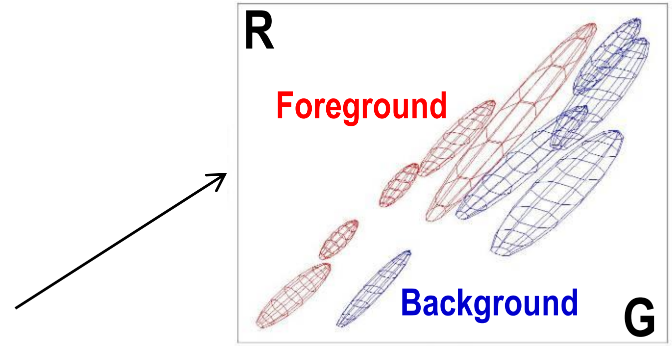


How to choose γ?

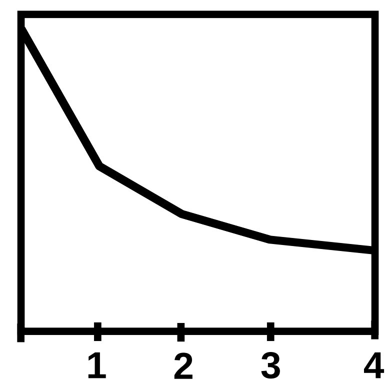


Iterated Graph Cuts

Result



Color model
(Mixture of Gaussians)



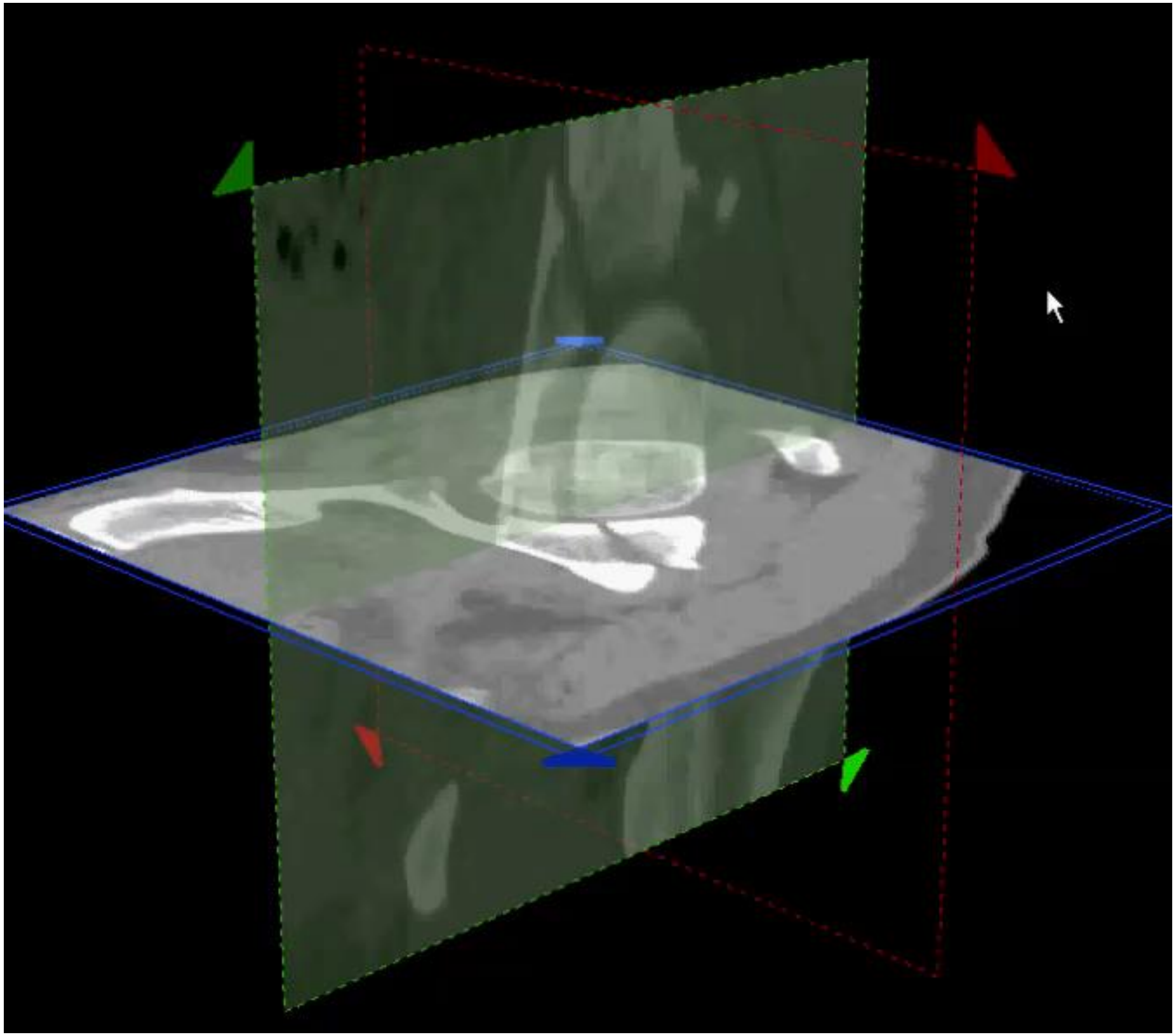
Energy after
each iteration

GrabCut: Example Results

- *This is included in all MS Office versions since 2010!*

Image source: Carsten Rother

Applications: Interactive 3D Segmentation



Slide credit: Yuri Boykov

[Y. Boykov, V. Kolmogorov, ICCV'03]

Summary: Graph Cuts Segmentation

- Pros
  - Powerful technique, based on probabilistic model (MRF).
  - Applicable for a wide range of problems.
  - Very efficient algorithms available for vision problems.
  - Becoming a de-facto standard for many segmentation tasks.

- Cons/Issues
  - Graph cuts can only solve a limited class of models
    - Submodular energy functions
    - Can capture only part of the expressiveness of MRFs
  - Only approximate algorithms available for multi-label case

References and Further Reading

- More information on image segmentation can be found in Chapter 5 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
(draft version available online)